Terraform Task

Task Description:

Launch Linux EC2 instances in two regions using a single Terraform file.

main.tf:

```
provider "aws" {
  alias  = "use1"
  region = "us-east-1"
}

provider "aws" {
  alias  = "usw2"
  region = "us-west-2"
}

data "aws_ami" "east_linux" {
  provider    = aws.use1
  most_recent = true
  owners      = ["amazon"]

  filter {
    name   = "name"
    values = ["amzn2-ami-hvm-*-x86_64-gp2"]
  }
}

data "aws_ami" "west_linux" {
  provider    = aws.usw2
  most_recent = true
  owners      = ["amazon"]

  filter {
    name   = "name"
    values = ["amzn2-ami-hvm-*-x86_64-gp2"]
  }
}

resource "aws_instance" "east_instance" {
  provider      = aws.use1
```

```
  ami           = data.aws_ami.east_linux.id
  instance_type = "t3.micro"

  tags = {
    Name = "Linux-Instance-East"
  }
}

resource "aws_instance" "west_instance" {
  provider      = aws.usw2
  ami           = data.aws_ami.west_linux.id
  instance_type = "t3.micro"

  tags = {
    Name = "Linux-Instance-West"
  }
}
```

```
provider "aws" {
  alias  = "use1"
  region = "us-east-1"
}

provider "aws" {
  alias  = "usw2"
  region = "us-west-2"
}

data "aws_ami" "east_linux" {
  provider    = aws.use1
  most_recent = true
  owners      = ["amazon"]

  filter {
    name   = "name"
    values = ["amzn2-ami-hvm-*-x86_64-gp2"]
  }
}

data "aws_ami" "west_linux" {
  provider    = aws.usw2
  most_recent = true
  owners      = ["amazon"]

  filter {
    name   = "name"
    values = ["amzn2-ami-hvm-*-x86_64-gp2"]
  }
}

resource "aws_instance" "east_instance" {
  provider      = aws.use1
  ami           = data.aws_ami.east_linux.id
  instance_type = "t3.micro"

  tags = {
    Name = "Linux-Instance-East"
  }
}
"main.tf" 51L, 883B
```

Terraform init:

```
~ $ nano main.tf
~ $ terraform init
Initializing the backend...
Initializing provider plugins...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/aws v6.35.1

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
~ $ terraform plan
```

Terraform plan:

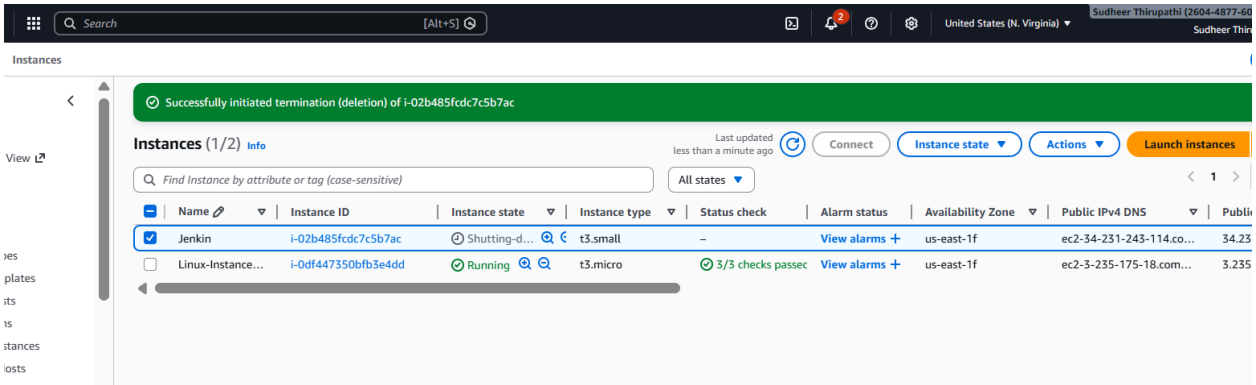
```
~ $ terraform plan
data.aws_ami.east_linux: Reading...
data.aws_ami.west_linux: Reading...
data.aws_ami.east_linux: Read complete after 0s [id=ami-05024c2628f651b80]
data.aws_ami.west_linux: Read complete after 1s [id=ami-0534a0fd33c655746]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # aws_instance.east_instance will be created
  + resource "aws_instance" "east_instance" {
      + ami                                  = "ami-05024c2628f651b80"
      + arn                                  = (known after apply)
      + associate_public_ip_address          = (known after apply)
      + availability_zone                    = (known after apply)
      + disable_api_stop                     = (known after apply)
      + disable_api_termination              = (known after apply)
      + ebs_optimized                        = (known after apply)
      + enable_primary_ipv6                  = (known after apply)
      + force_destroy                        = false
      + get_password_data                    = false
      + host_id                              = (known after apply)
      + host_resource_group_arn              = (known after apply)
      + iam_instance_profile                 = (known after apply)
      + id                                   = (known after apply)
      + instance_initiated_shutdown_behavior = (known after apply)
      + instance_lifecycle                   = (known after apply)
      + instance_state                       = (known after apply)
      + instance_type                        = "t3.micro"
      + ipv6_address_count                   = (known after apply)
      + ipv6_addresses                       = (known after apply)
      + key_name                             = (known after apply)
      + monitoring                           = (known after apply)
      + outpost_arn                          = (known after apply)
      + password_data                        = (known after apply)
      + placement_group                      = (known after apply)
      + placement_group_id                   = (known after apply)
      + placement_partition_number           = (known after apply)
```

Terraform apply:

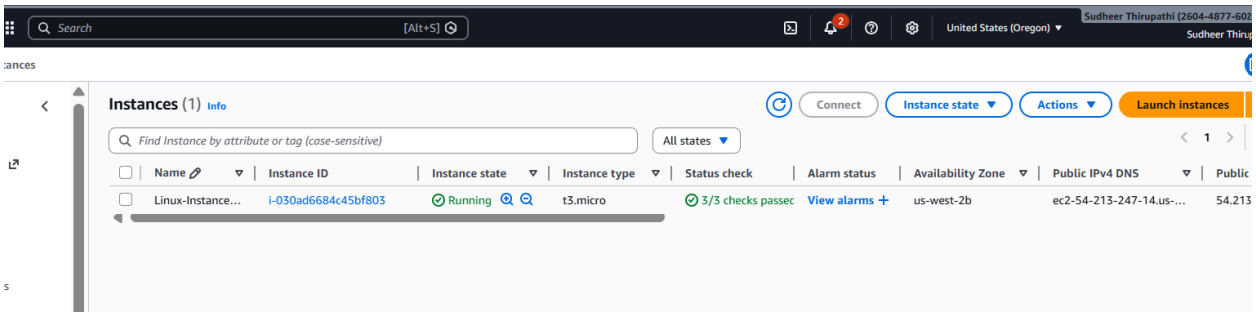
```
~ $ terraform apply
data.aws_ami.east_linux: Reading...
data.aws_ami.east_linux: Read complete after 0s [id=ami-05024c2628f651b80]
data.aws_ami.west_linux: Reading...
data.aws_ami.west_linux: Read complete after 1s [id=ami-0534a0fd33c655746]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # aws_instance.east_instance will be created
  + resource "aws_instance" "east_instance" {
      + ami                                  = "ami-05024c2628f651b80"
      + arn                                  = (known after apply)
      + associate_public_ip_address          = (known after apply)
      + availability_zone                    = (known after apply)
      + disable_api_stop                     = (known after apply)
      + disable_api_termination              = (known after apply)
      + ebs_optimized                        = (known after apply)
      + enable_primary_ipv6                  = (known after apply)
      + force_destroy                        = false
      + get_password_data                    = false
      + host_id                              = (known after apply)
      + host_resource_group_arn              = (known after apply)
      + iam_instance_profile                 = (known after apply)
      + id                                   = (known after apply)
      + instance_initiated_shutdown_behavior = (known after apply)
      + instance_lifecycle                   = (known after apply)
      + instance_state                       = (known after apply)
      + instance_type                        = "t3.micro"
      + ipv6_address_count                   = (known after apply)
      + ipv6_addresses                       = (known after apply)
      + key_name                             = (known after apply)
      + monitoring                           = (known after apply)
      + outpost_arn                          = (known after apply)
      + password_data                        = (known after apply)
      + placement_group                      = (known after apply)
      + placement_group_id                   = (known after apply)
      + placement_partition_number           = (known after apply)
      + primary_network_interface_id         = (known after apply)
```

```
      + vpc_security_group_ids               = (known after apply)
      + cpu_options (known after apply)

      + ebs_block_device (known after apply)

      + enclave_options (known after apply)

      + ephemeral_block_device (known after apply)

      + instance_market_options (known after apply)

      + maintenance_options (known after apply)

      + metadata_options (known after apply)

      + network_interface (known after apply)

      + primary_network_interface (known after apply)

      + private_dns_name_options (known after apply)

      + root_block_device (known after apply)

      + secondary_network_interface (known after apply)
    }

Plan: 2 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

aws_instance.east_instance: Creating...
aws_instance.west_instance: Creating...
aws_instance.east_instance: Still creating... [00m10s elapsed]
aws_instance.west_instance: Still creating... [00m10s elapsed]
aws_instance.east_instance: Creation complete after 12s [id=i-0df447350bfb3e4dd]
aws_instance.west_instance: Creation complete after 13s [id=i-030ad6684c45bf803]

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.
~ $
```

Us-east-1:



Us-west-2:



Destroy:

```
~ $ terraform destroy
data.aws_ami.east_linux: Reading...
data.aws_ami.west_linux: Reading...
data.aws_ami.east_linux: Read complete after 0s [id=ami-05024c2628f651b80]
aws_instance.east_instance: Refreshing state... [id=i-0df447350bfb3e4dd]
data.aws_ami.west_linux: Read complete after 0s [id=ami-0534a0fd33c655746]
aws_instance.west_instance: Refreshing state... [id=i-030ad6684c45bf803]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  - destroy

Terraform will perform the following actions:

  # aws_instance.east_instance will be destroyed
  - resource "aws_instance" "east_instance" {
      - ami                                  = "ami-05024c2628f651b80" -> null
      - arn                                  = "arn:aws:ec2:us-east-1:260448776023:instance/i-0df447350bfb3e4dd" -> null
      - associate_public_ip_address          = true -> null
      - availability_zone                    = "us-east-1f" -> null
      - disable_api_stop                     = false -> null
      - disable_api_termination              = false -> null
      - ebs_optimized                        = false -> null
      - force_destroy                        = false -> null
      - get_password_data                    = false -> null
      - hibernation                          = false -> null
      - id                                   = "i-0df447350bfb3e4dd" -> null
      - instance_initiated_shutdown_behavior = "stop" -> null
      - instance_state                       = "running" -> null
      - instance_type                        = "t3.micro" -> null
      - ipv6_address_count                   = 0 -> null
      - ipv6_addresses                       = [] -> null
      - monitoring                           = false -> null
      - placement_partition_number           = 0 -> null
      - primary_network_interface_id         = "eni-06ee9192e24fa36e4" -> null
      - private_dns                          = "ip-172-31-69-209.ec2.internal" -> null
      - private_ip                           = "172.31.69.209" -> null
      - public_dns                           = "ec2-3-235-175-18.compute-1.amazonaws.com" -> null
      - public_ip                            = "3.235.175.18" -> null
```

```
      - root_block_device {
          - delete_on_termination = true -> null
          - device_name           = "/dev/xvda" -> null
          - encrypted             = false -> null
          - iops                  = 100 -> null
          - tags                  = {} -> null
          - tags_all              = {} -> null
          - throughput            = 0 -> null
          - volume_id             = "vol-00011d39bbbfa5d54" -> null
          - volume_size           = 8 -> null
          - volume_type           = "gp2" -> null
            # (1 unchanged attribute hidden)
        }
    }

Plan: 0 to add, 0 to change, 2 to destroy.

Do you really want to destroy all resources?
  Terraform will destroy all your managed infrastructure, as shown above.
  There is no undo. Only 'yes' will be accepted to confirm.

  Enter a value: yes

aws_instance.east_instance: Destroying... [id=i-0df447350bfb3e4dd]
aws_instance.west_instance: Destroying... [id=i-030ad6684c45bf803]
aws_instance.east_instance: Still destroying... [id=i-0df447350bfb3e4dd, 00m10s elapsed]
aws_instance.west_instance: Still destroying... [id=i-030ad6684c45bf803, 00m10s elapsed]
aws_instance.east_instance: Still destroying... [id=i-0df447350bfb3e4dd, 00m20s elapsed]
aws_instance.west_instance: Still destroying... [id=i-030ad6684c45bf803, 00m20s elapsed]
aws_instance.east_instance: Still destroying... [id=i-0df447350bfb3e4dd, 00m30s elapsed]
aws_instance.west_instance: Still destroying... [id=i-030ad6684c45bf803, 00m30s elapsed]
aws_instance.east_instance: Still destroying... [id=i-0df447350bfb3e4dd, 00m40s elapsed]
aws_instance.west_instance: Still destroying... [id=i-030ad6684c45bf803, 00m40s elapsed]
aws_instance.west_instance: Destruction complete after 40s
aws_instance.east_instance: Still destroying... [id=i-0df447350bfb3e4dd, 00m50s elapsed]
aws_instance.east_instance: Still destroying... [id=i-0df447350bfb3e4dd, 01m00s elapsed]
aws_instance.east_instance: Still destroying... [id=i-0df447350bfb3e4dd, 01m10s elapsed]
aws_instance.east_instance: Still destroying... [id=i-0df447350bfb3e4dd, 01m20s elapsed]
aws_instance.east_instance: Destruction complete after 1m20s

Destroy complete! Resources: 2 destroyed.
~ $
```